1. GROUP NUMBER AND GROUP MEMBERS

Group 4
Members: CHUNG JIA CHENG, TJONG JIA HUI KENNETH, TOH EU LEONG, YEUNG HAN JIE, IVY SIOW XUAN ZI

2. TITLE OF GROUP PROJECT

The title should be **no more than 80** characters (including space). The title should be descriptive and specific rather than general.

Autonomous Agentic System for Market Intelligence Analysis

3. ABSTRACT OF THE PROPOSAL OF GROUP PROJECT

Please provide a succinct description of the proposal of the group project in **no more than** 300 words. The abstract must capture the academic significance of the research, the problem statement to be addressed, the approach and feasibility of the study.

a) Novelty of Group Project

Currently, many generative ai projects implemented for an enterprise use case are centered around RAG systems like knowledge bases or an agentic document workflow. These systems largely use unstructured data to extract and provide actionable insights but largely do not autonomously disseminate the work. We aim to incorporate a multi-modal strategy by ingesting structured and unstructured data whose contexts' can be combined to form a larger picture of a business competitiveness problem to fix, and ideally for the agent to be able to finally disseminate to executives, quantitative probabilities and confidence levels behind the agentic system's recommendations so as to assist their decision-making using typesafe's System One model Jev.

b) Problem Statements to be Addressed

As it stands, most generative ai solutions centered around data science workflows stop at the insights garnered and qualitative descriptions and recommendations. There are currently no clear methods to quantitatively justify a model's output to be used for executive decision making other than a human analyst to qualitatively contextualise the results for them. With jev, clear probabilistic indicators can be used to aid executive understanding for them to decide what is the best action to take.

c) Methodology/ Approach and Feasibility of study

We are looking to make a minimum viable product of a multi-agent system that utilises a knowledge base and a structured OLTP tabular database to conduct market intelligence analysis, and we will use 5 real business scenarios with detailed contexts to test whether the agentic system makes the correct recommendations.

4. DETAILS OF PROPOSAL

Give a description of the group project in no more than **3 pages (Arial font size 10, single line spacing)**. Information shall include specific project aims, gaps and challenges to be solved, problem statements and significance, system to be developed in the group project, Innovation and novelty of the group project. Information provided should be sufficiently self-contained without need for further reference to other materials. Please attach diagrams, plots, tables and other supporting visuals.

(a) Specific project aims and

We will design a multi-agent system that conducts market intelligence research and ingests relevant relational or non-relational data to specialized agents to work on, whose outputs can then be used by a new model recently released to provide quantitative insights, statistics and confidence levels behind this decision model. A LLM is used instead to provide the rationale behind the decision model's choice.

To build it, we will:

1. Ingest business data. But for our cases as we do not have access to commercial dataset, we will use Kaggle datasets for structured tabular analyses. For unstructured data sources, we scrape news articles and generate artificial business rules and SOP documents for unstructured data.
2. We initialize four agents to conduct market intelligence analysis: Competitive Intelligence Agent, Customer Intelligence Agent, Product Intelligence and Market Analyst Agent
3. For agents requiring structured CSV files/Database dumps, we load them into OLTP databases, and for agents requiring unstructured textual data, we perform chunking, embedding and load them into vector store like milvus or pinecone or qdrant.
4. We then take in a user query. If the query requires structured data present in a OLTP SQL database, we use a text-to-sql tool to generate tabular query results. If the query requires using unstructured data like document files or any other textual sources, we perform RAG over the document knowledge base using the vector store initialized.
5. We store the extracted outputs in a shared state and pass it to the jev decision model to output exact probabilities and confidence levels for a set of actions that needs to be taken to resolve the highlighted issues with market competitiveness.
6. LLM summarizes the rationale behind the decision model's calculations and quantitative analysis, and we establish human-in-the-loop guardrails to ensure that a human verifies the analysis and make the appropriate decision. To do this, we ensure that the agents always provide attribution to why and how they came to their analytic conclusions.
7. We then deploy this service as a docker container on cloud infrastructure and perform user testing and agentic benchmarking like tool call rate, task completion rate etc.

(b) Gaps and challenges to be solved

Firstly, there is a gap between quantitative analyses using classical data science techniques and qualitative explanation for complex decisions and issues found. Executives cannot know what is behind a LLM's thinking process to decide whether to trust it or not.

Furthermore, the largest bottleneck in data science workflows is in performing data cleaning, feature engineering and outlier analysis. It is estimated that up to 80% of the time spent is used for this single important step. A crucial step that developers are bound to not completely do it accurately may end up accumulating losses for the enterprise.

Finally, there is the issue of data siloing because it leads to isolated data sources that may not corroborate nor support each other in tackling issues such as stifled collaboration and operational inefficiencies from working with incomplete and inaccurate data.

(c) Problem statements and significance

We want to check if superior coding agents, can help to cut down the time spent on query development, data cleaning and feature engineering, and whether a human-in-the-loop guardrail for human analysts to verify and discern the agentic system's outputs help to aid efficiency. Knowing the answer to this can help us decide whether the costs of implementing a new agentic analytics system is worth the long-term productivity gains from analytical insights and patterns.

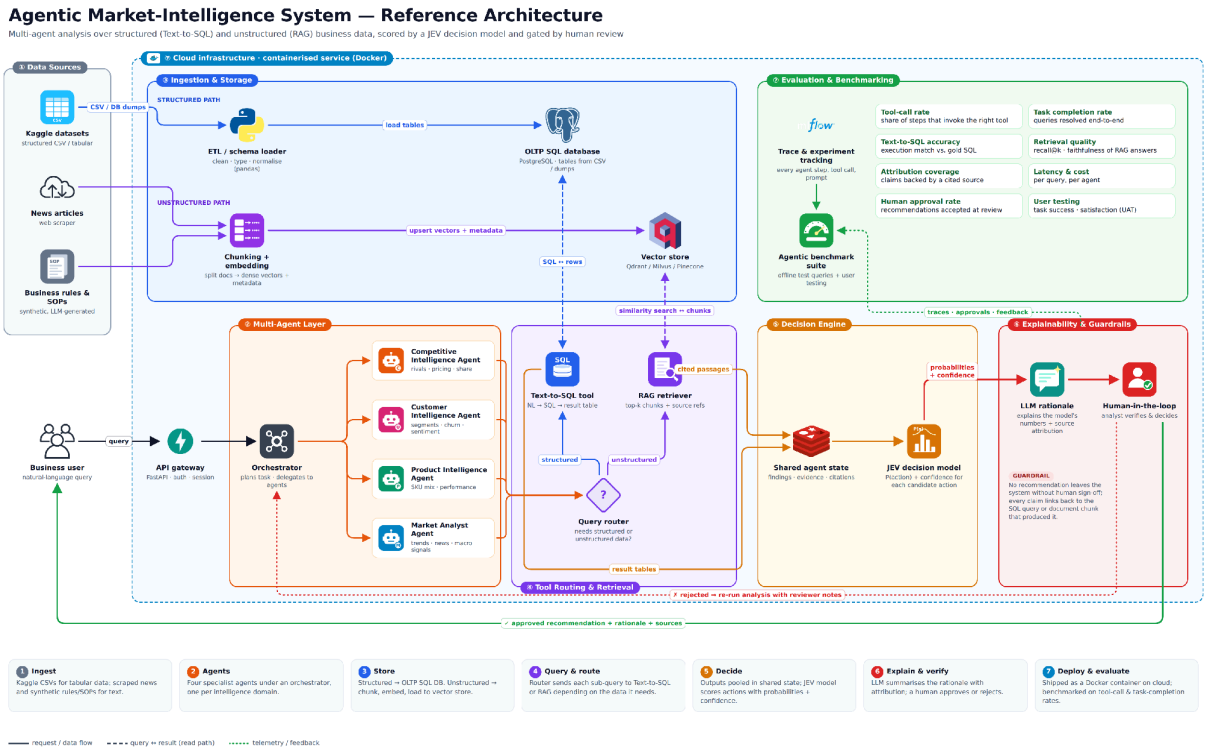
We also want to check if unstructured and structured data's contexts can be consolidated to form a better unified view and to see if it can hence provide better insights of how well the enterprise is doing internally and compared to its market competitors, and whether better action recommendations can be produced to aid their business processes.

Lastly we aim to experiment how well novel decision models like JEV fare in making good business decisions as compared to executive's business acumen. If we can provide good quantitative metrics as to why our system chose a certain action to recommend over the others, it greatly aids in making explainable AI to non-technical audiences, and it aids the detection and debugging of silent logic errors for production maintenance.

(e) Innovation and novelty of the group project

The innovation in our project comes not just from using a novel decision-based model but also to integrate structured and non-structured data into a unified source of truth to aid business decision-making using clean and aggregated data sources. The JEV model is used at the last stage to present quantitative probabilistic and confidence metrics that business executives can look to with confidence. JEV runs at a much lower cost than frontier LLMs, so this is a opportunity for cost reduction.

(d) System to be developed in the group project



System diagram showing how this might come together subject to feasibility constraints.